## 1. Approach, Assumptions and Solution

**Approach:** Organised dealership CSV and Excel files into source-specific folders. Inspected column names, data types and sample records before selecting five core fields: Date, Dealership, Department, Rating and Comments. Standardised inconsistent schemas and consolidated usable records into a single dataset for analysis.

Validated dates, ratings and departments, including manual inspection of problematic Excel entries. Applied PII masking before text analysis. Developed an interactive dashboard to explore customer feedback by dealership, department, rating, date and keyword.

**Assumptions and Limitations:** Three dealerships were excluded to simplify the initial analysis. Findings therefore do not represent all 24 dealerships. Customer feedback was treated as reported experience rather than verified fact. Keyword matches indicate potential issues, not confirmed incidents. Inconsistent survey collection may introduce reporting bias, and feedback counts alone cannot establish changes in complaint rates.

## 2. Technology Choices and Rationale

- **Python / Pandas:** Consolidation, cleaning, validation and standardisation across heterogeneous files.
- **VS Code / Excel Viewer Extensions:** Manual inspection of source files, column structures, formatting inconsistencies and problematic records.
- **NLTK:** Stopword removal and lemmatisation to standardise text for keyword frequency analysis.
- **Keyword Matching and Verification:** Interactive keyword selection filters feedback by relevant words, allowing users to inspect matching comments and star ratings. Original comments support manual verification of identified issues. Future rule-based exclusions could remove irrelevant matches from visualisations.
- **Plotly / Dash:** Interactive exploration of rating trends, keyword frequencies and customer feedback.
- **No External LLMs:** Avoided due to the risk of residual PII in customer comments, even after masking. Prioritised local, transparent and reproducible analysis.

## 3. Data Quality Observations

- **Inconsistent Schemas:** Different column names, formats and structures across dealership exports.
- **Date Integrity:** Mixed DD/MM and MM/DD conventions, Excel serial dates, incorrect ordering and ambiguous entries.
- **Rating Quality:** Inconsistent numeric and text-based representations requiring sanitisation.
- **Duplicate and Invalid Records:** Repeated exports, missing values and irregular entries requiring review.
- **Privacy Risks:** Customer comments containing personally identifiable information (PII).
- **Inconsistent Feedback Timing:** Feedback collected at different stages of the customer journey, including while awaiting service, waiting for callbacks and after service completion. This makes satisfaction ratings less comparable.
- **Collection Bias:** Inconsistent reporting periods, survey timing and collection practices may distort customer satisfaction trends.

**Key Finding:** The primary challenge is not the absence of customer feedback, but the lack of consistent data collection, validation and governance. A standardised reporting process is required for reliable, scalable customer experience monitoring.

# 4. Customer Experience Findings

## 4.1 Waiting Times and Communication




Figure 1: Monthly feedback mentioning waiting times or communication delays.

**Trend Analysis:** Waiting times and communication delays are recurring concerns, predominantly within Service (353 of 492 keyword-matched records, 71.7%) and across multiple dealerships. Low-rated matches persist throughout the reporting period, suggesting a potential network-wide operational issue.

## 4.2 Pricing and Invoicing

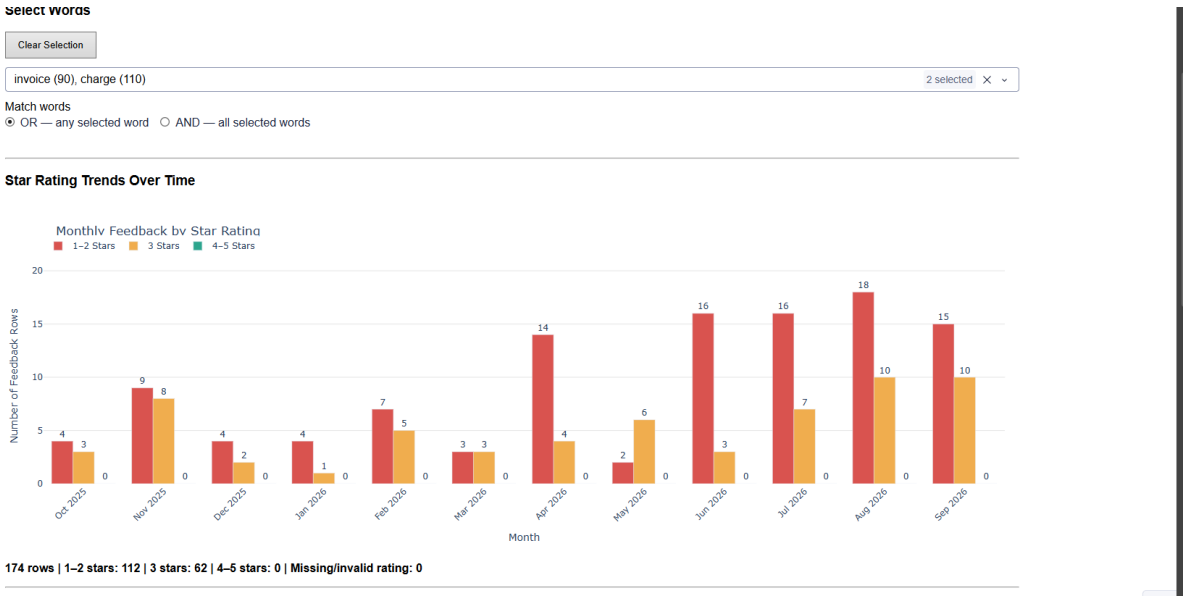



Figure 2: Monthly feedback mentioning invoices or charges.

**Trend Analysis:** Of 174 keyword-matched comments, 112 (64.4%) were rated 1–2 stars. Most originated in Service (132; 75.9%), spanning 19 dealerships. Matches increased during June–September

2026.

**Limitation:** Keyword matches do not confirm overcharging. Changes in feedback volume prevent conclusions about increasing complaint rates.

### 4.3 Booking System

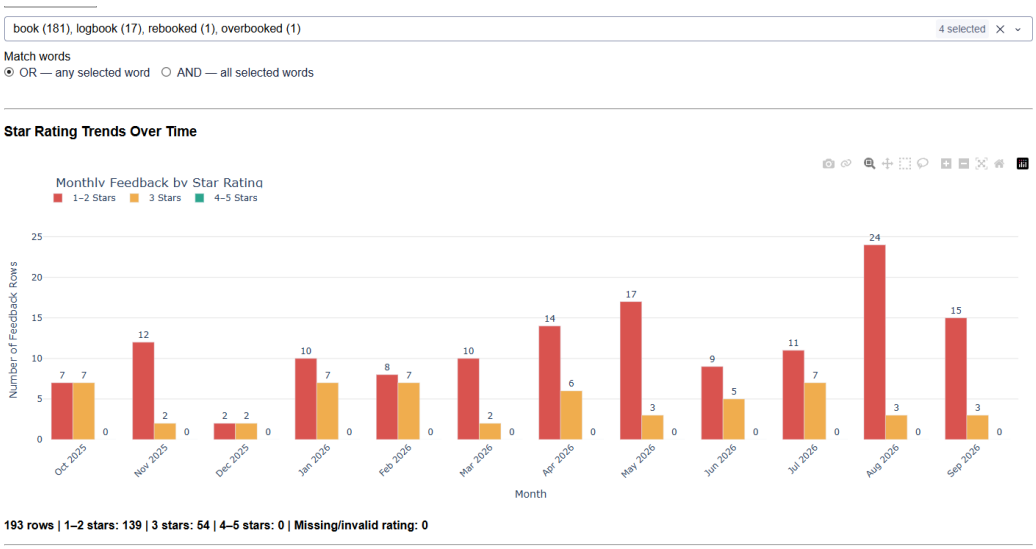



Figure 3: Monthly booking-related feedback and star ratings.

**Trend Analysis:** Of 193 booking-related keyword matches, 139 (72%) were rated 1–2 stars. Most occurred in Service (95%) across 18 dealerships. Matches increased following the May 2026 booking system launch, peaking in August.

**Interpretation:** The timing suggests potential adoption or usability issues, although a causal relationship has not been established. Complaint rates cannot be inferred without accounting for total feedback volume.

**NLP Validation and Limitations:** The dashboard allows users to inspect all comments and ratings matching selected keywords, supporting manual verification. However, keyword matching may include unrelated feedback or miss complaints expressed differently. Future improvements could introduce exclusion rules to filter irrelevant matches from visualisations. Classification accuracy has not been formally measured.

## 5. Recommendations and Business Benefits

#### Immediate Priority: Booking and Service Experience

Investigate the new booking system introduced in May 2026, alongside service workflows, communication and invoicing processes. These three areas account for the recurring issues identified in customer feedback:

- **Booking:** Improve usability, booking reliability and customer confirmation processes.
- **Waiting and Communication:** Improve appointment scheduling, service updates and callback processes.
- **Pricing and Invoicing:** Improve quote transparency, customer authorisation and invoice accuracy.

**Business Value:** Addressing these connected issues could reduce failed bookings, unnecessary waiting, repeat enquiries and billing disputes. Expected benefits include improved customer satisfaction, stronger retention, reduced administrative workload and more efficient service operations.

| Recommendation | Benefit to the Company |
|---|---|
| Improve booking system | Fewer booking failures, reduced waiting times and improved customer satisfaction. |
| Standardise processes across dealerships | Consistent customer experience and easier identification of underperforming dealerships. |
| Standardise data collection | Higher-quality data, more reliable insights and better process improvements. |
| Automate data synchronisation | Eliminate manual Excel submissions, reduce human error and save staff time. |
| Implement robust data validation | Prevent reporting failures caused by missing columns, schema changes or invalid records. |
| Strengthen PII governance | Restrict sensitive data access, reduce data leakage risk and potential legal liabilities. |
| Use pseudonymous customer IDs | Enable customer journey and retention analysis without unnecessarily exposing PII. |
| Collect independent customer feedback | Reduce potential staff influence and improve accuracy of customer satisfaction measurements. |
| Maintain documentation and staff training | Easier onboarding, system continuity and reduced dependence on individual employees. |
| Connect bookings, transactions and feedback | Identify recurring issues, track customer retention and prioritise improvements based on business impact. |

**Overall Benefit:** A more reliable, secure and automated system that reduces operational costs, improves customer experience and enables continuous, data-driven process improvement.

## 6. AI Usage and Review

- **Code Generation:** Used ChatGPT to generate Pandas code for data cleaning, validation, Excel formulas and sanitisation. Reviewed and tested generated code before implementation.
- **Data Consolidation:** Used AI-assisted coding to combine heterogeneous Excel and CSV files into one DataFrame. Implemented logging to identify successful and failed imports, iteratively debugging until the consolidation process worked.
- **MVP Development:** Used AI tools to develop an interactive data analysis MVP, demonstrated through the visualisations in this report.
- **Verification:** Cross-checked outputs against original source files, inspected error logs and manually resolved problematic dates, column mappings and inconsistent values.
- **Privacy and NLP:** Avoided external LLM-based feedback classification due to residual PII risks. Used NLTK and keyword matching for transparent, reproducible analysis.

**Overall:** AI accelerated coding, consolidation and MVP development, while human review, error logging and iterative testing supported the reliability of the final analysis.